

# A Nature Astronomy-inspired manuscript in L<sup>A</sup>T<sub>E</sub>X

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This unofficial template is intended for writing manuscripts, notes and polished preprints in a visual style inspired by Nature Astronomy. The current setup combines Palatino-style serif text and matching mathematics with compact two-column text, REVTeX-inspired section headings, author–year citations and concise captions. A title, author and abstract block sits above both columns. Journal production metadata such as received, accepted or publication dates, DOI information and volume details are omitted.

A manuscript can begin directly with the scientific narrative rather than with a heading labelled “Introduction”. The opening paragraphs should establish the context, identify the unresolved problem and state the contribution succinctly. The selected `authoryear` option enables author–year citations through `natbib`; use `\citep` for parenthetical citations (W. Dong & A. Melatos, 2024, 2025; W. Dong et al., 2026) and `\citet` for textual citations, as in W. Dong & A. Melatos (2024). The reference list is generated from `examples.bib` with the `aasjournalv7` BibTeX style.

This file is deliberately a clean authoring template rather than a simulation of the journal’s production workflow. Its purpose is to make drafts pleasant to read while keeping the source conventional and easy to adapt later.

The `authoryear` option enables author–year citations; omitting it gives numerical superscripts. Use `natureheadings` for the default heading style. To apply REVTeX styling through a chosen level, set `revtexheadings=` to section, subsection or subsubsection. The shorthand `revtexheadings` styles all three

levels.

## I. RESULTS

### A compact two-column hierarchy

The principal body is set in two columns. Sections have centred, bold, uppercase headings with Roman numerals. Subsections remain left-aligned and bold, and subsubsections left-aligned and italic, both without displayed numbers. Text and mathematics use matching Palatino-style fonts. A simple polynomial illustrates a single-column equation:

$$f(x) = ax^2 + bx + c, \quad (1)$$

where  $a$ ,  $b$  and  $c$  are constant coefficients.

For an equation that needs both columns, the `widetext` environment interrupts the two-column flow with a full-width display:

$$\begin{aligned} I &= \int_{x_0}^{x_1} f(x) \, dx = \int_{x_0}^{x_1} (ax^2 + bx + c) \, dx, \\ &= \left[ \frac{a}{3}x^3 + \frac{b}{2}x^2 + cx \right]_{x_0}^{x_1} = \frac{a}{3}(x_1^3 - x_0^3) + \frac{b}{2}(x_1^2 - x_0^2) + c(x_1 - x_0). \end{aligned} \quad (2)$$

Table 1. Illustrative polynomial coefficients

| Example | $a$ | $b$ | $c$ |
|---------|-----|-----|-----|
| A       | 1   | 0   | 0   |
| B       | 1   | 2   | 1   |
| C       | 2   | −1  | 3   |

Here  $I$  is the definite integral of  $f(x)$  between  $x_0$  and  $x_1$ . Text resumes in two columns after equation 2; the short rules mark the transition into and out of the wide display.

Figure 1 illustrates the default single-column treatment. For a figure spanning the whole page width, use the standard `figure*` environment.

## Tables

Tables use compact text and simple horizontal rules provided by `booktabs`. The example lists illustrative coefficients for the polynomial above; use `table*` for a table spanning both columns.

## II. DISCUSSION

Use the layout as a writing environment for a manuscript, internal draft, research note or preprint. The source remains ordinary L<sup>A</sup>T<sub>E</sub>X, so the manuscript can later be transferred to a journal-specific

template without restructuring the content.

## METHODS

Methods can continue in the same two-column format. For a real manuscript, place sufficient implementation detail here to reproduce the analysis, together with statistical conventions, software versions and selection criteria.

## DATA AVAILABILITY

State where the data underlying the study can be obtained, if this section is useful for the document.

## CODE AVAILABILITY

State where analysis code is archived, if applicable.



**Figure 1. Example single-column figure.** Replace the framed placeholder with `\includegraphics`. Captions should explain symbols and uncertainties sufficiently for the figure to be interpreted without repeatedly returning to the main text.

## ACKNOWLEDGEMENTS

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## AUTHOR CONTRIBUTIONS

F.A. conceived the study. F.A. and S.A. performed the analysis. All authors discussed the results and contributed to the manuscript.

## COMPETING INTERESTS

The authors declare no competing interests.

## REFERENCES

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- Dong, W., Melatos, A., O'Neill, N. J., Meyers, P. M., & Boek, D. K. 2026, Measuring the Crust-Superfluid Coupling Time-Scale for 105 UTMOST Pulsars with a Kalman Filter, *Monthly Notices of the Royal Astronomical Society*, 545, staf2053, doi: [10.1093/mnras/staf2053](https://doi.org/10.1093/mnras/staf2053)

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